



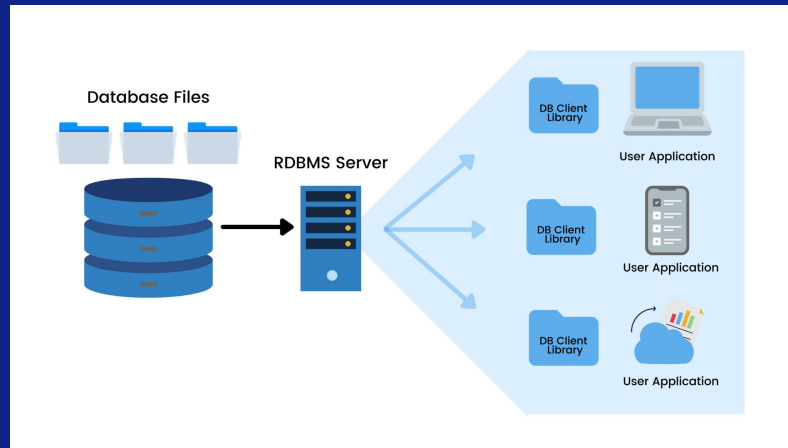
SUMMARIZER

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RDBMS

A relational database management system (RDBMS) is a program that allows you to create, update, and administer a relational database. A relational database is a type of database that stores and organizes data points with defined relationships for quick access. Most relational database management systems use the SQL language to access the database.





DATA BASE

Benefits

They allow you to store large amounts of data efficiently and securely.

They facilitate the sharing and integration of data between different applications and users.

Improve data quality and consistency by preventing redundancies and errors.

Disadvantages

Require a high cost to acquire, maintain and update the necessary hardware and software.

Require specialized technical knowledge to design, implement and administer the bases



Relational vs Non-relational

Feature	Non Relational Database	Relational Database
Availability	Good	Good
Consistency	Poor	Good
Data Storage	Optimised for Huge Data	Medium - Large Data Size
Performance	High	Low
Reliability	Poor	Good
Scalability	High	High (but more expensive)

- relational is a type of database that uses tables, fields, and columns to organize data.
- non-relational is a type of database that does not use this structure and stores data in a hierarchical form



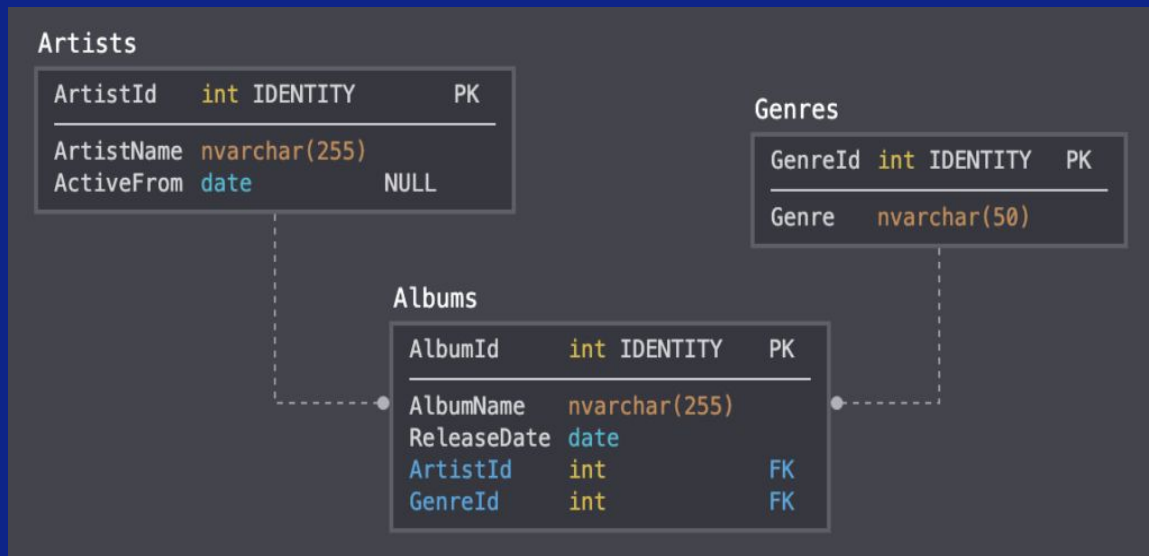
MySQL is a relational database that supports SQL and has high performance, reliability and ease of use.

It is used by applications such as Facebook, Twitter, Netflix, Uber and Airbnb



Database Relations

A database relationship is an association between two or more tables that contain related data. Tables are structures that store data in rows and columns.





Primary Key and Foreign Keys

- Primary key: is the key that uniquely identifies each record in a table and cannot be null.
- Foreign key: is the key that references the primary key of another table and establishes a relationship between them

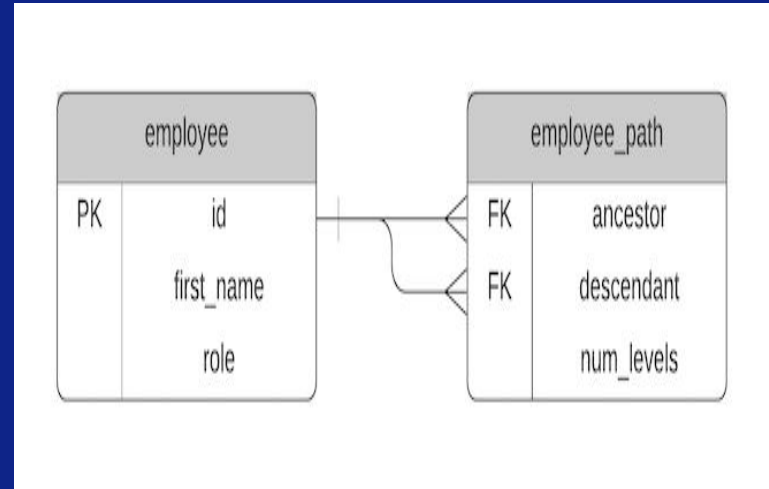


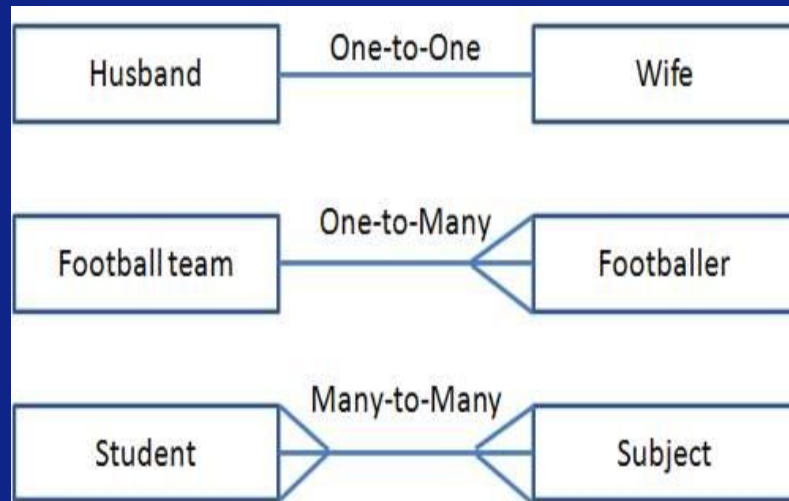


Table Relationships

one-to-one is a link between two tables, where each record in each table appears only once

one-to-many is a link between two tables, where each record in one table can be related to multiple records in the other table, but each record in the other table can only be related to one record in the first table

many-to-many is a link between two tables, where each record in one table can be related to several records in the other table and vice versa



CRUD

**Create,
Read,
Update,
Delete**

These are the four basic operations that we can perform with most traditional database systems and are the basis for interacting with any database.





Let the Bolo Rei Begin

- SELECT - extract data from a database
- UPDATE - updates data in a database
- DELETE - deletes data from a database
- INSERT INTO - insert new data into a database
- CREATE DATABASE - creates a new database
- ALTER DATABASE - modify a database
- CREATE TABLE - creates a new table
- ALTER TABLE - modify a table
- JOIN - combines rows from two or more tables based on a common condition
- GROUP BY - groups rows that have the same values in one or more columns
- HAVING - filters groups created by the GROUP BY command
- ORDER BY - sorts the result of a query in ascending or descending order
- DISTINCT - eliminates duplicate values in the result of a query
- UNION - combines the result of two or more queries into a single result set



But the most important

How to start mysql in terminal??

```
mysql.server -u root localhost
```

???

I don't know





Thank You

